

§ Lecture 16.0

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Handling a bit of infinity.

We already have defined that for set E with no upper bound

$$\sup(E) = +\infty$$

Similarly $\sup \emptyset = -\infty$.

Issues:

- ① We can't compare them yet!
- ② If we want to define $\sup \{ \sup E_i \}$, then we run into trouble as $\sup$ is only defined for set of real numbers. and $\pm\infty$ are not real number.

Extended real number system.

$$\mathbb{R}^* = \mathbb{R} \cup \{+\infty, -\infty\}$$

A real number x is called finite if $x \in \mathbb{R}$ otherwise infinite.

Negation of extended reals. Let $x \in \mathbb{R}^*$ then if $x \in \mathbb{R}$, $-x$ is usual negation. Further $-(+\infty) = -\infty$, $-(-\infty) = +\infty$.

We don't define any other algebraic operation on $\pm\infty$.

Ordering of extended reals. Let $x, y \in \mathbb{R}^*$. We say that $x \leq y$

iff One of the following three statements is true.

- (i) $x, y \in \mathbb{R}$ and $y - x \geq 0$. [Usual]
- (ii) $x = -\infty$
- (iii) $y = +\infty$

That is $-\infty < y$ for any $y \in \mathbb{R}^*$. In particular $-\infty \leq +\infty$.

Distinct, so $-\infty < +\infty$.

Supremum of the subsets of the extended reals.

Let $E \subseteq \mathbb{R}^*$. $\sup(E)$ is defined as follows.

- (a) If $E \subseteq \mathbb{R}$, then $\sup(E)$ is usual definition. [$+\infty, -\infty$ can appear as answer]
- (b) If $+\infty \in E$ then $\sup(E) = +\infty$.
- (c) If $+\infty \notin E$ and $-\infty \in E$, then $\sup(E) = \sup(E - \{-\infty\})$ [on usual $\mathbb{R}$]

Similarly $\inf(E) = -\sup(-E)$.

Suprema and infima of sequences.

Let $(a_n)_{n \geq 1}$ be a sequence of real numbers. Then

$$\sup(a_n)_{n \geq 1} \equiv \sup a_n := \sup \{ a_n : n \geq 1 \}$$

$$\inf(a_n)_{n \geq 1} \equiv \inf a_n := \inf \{ a_n : n \geq 1 \}.$$

Monotone sequences. Let $(a_n)_{n \geq 1}$ be a sequence. It is said to be

- (i) Monotone increasing if $a_{n+1} \geq a_n \quad \forall n \geq 1$.
- (ii) Monotone decreasing if $a_{n+1} \leq a_n \quad \forall n \geq 1$

We say that a sequence is monotone if it is either monotone increasing or monotone decreasing.

Lemma. (Monotone bounded sequences converge). Let $(a_n)_{n \geq 1}$ be a sequence of real numbers.

- (i) If $(a_n)_{n \geq 1}$ is monotone increasing and upper bounded by $M_0 \in \mathbb{R}$, then

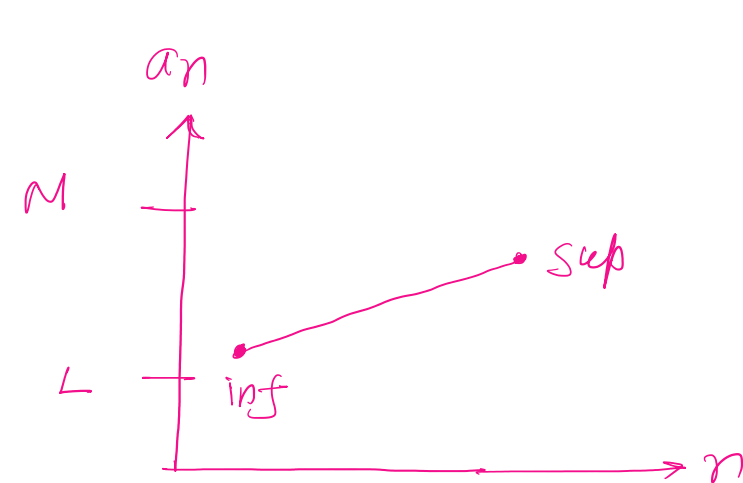
$$\lim_{n \rightarrow \infty} a_n = \sup_{n \geq 1} a_n \leq M_0 \quad [(a_n)_{n \geq 1} \text{ is convergent}]$$

- (ii) If $(a_n)_{n \geq 1}$ is monotone decreasing and lower bounded by $M_1 \in \mathbb{R}$, then

$$\lim_{n \rightarrow \infty} a_n = \inf_{n \geq 1} a_n \geq M_1.$$

Proof. ① Given $a_{n+1} \geq a_n \quad \forall n \geq 1$

$$a_n \leq M \quad \forall n \geq 1$$



Since we don't know the limit exists we can't take limit both sides.

Let $S = \sup_n a_n$. Our goal is to show that $(a_n)_{n \geq 1}$ converges to S .

Fix $\varepsilon > 0$.

Note that $S \leq M$.

Since any upper bound on $(a_n)_{n \geq 1}$ must be larger than S , $S - \varepsilon$ can't be an upper bound.

$$\Rightarrow \exists N \geq 1 \quad \text{s.t.} \quad a_N > S - \varepsilon.$$

$$\text{But } a_{n+1} \geq a_n > S - \varepsilon. \text{ That is } \forall n \geq N \quad a_n > S - \varepsilon.$$

Now $\forall n \geq N$

$$S - \varepsilon < a_n \leq S < S + \varepsilon$$

$$\Rightarrow |a_n - S| < \varepsilon$$

That is $(a_n)_{n \geq 1}$ is convergent and $\lim_{n \rightarrow \infty} a_n = \sup_n(a_n) \leq M$.

- (ii) Let $\inf_n a_n = L \geq M_1$.

Fix $\varepsilon > 0$. Then since every lower bound is smaller than L , $L + \varepsilon$

can't be a lower bound. That is $\exists N \geq 1 \quad \text{s.t.}$

$$a_N < L + \varepsilon \quad (\text{Then } a_{n+1} \leq a_N < L + \varepsilon)$$

Then $\forall n \geq N$

$$L - \varepsilon < L \leq a_n < L + \varepsilon \Rightarrow |a_n - L| < \varepsilon$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = \inf_{n \geq 1} a_n.$$

Example: Let $0 < x < 1$. Then $\lim_{n \rightarrow \infty} x^n = 0$.

Solution: Take the sequence $(x^n)_{n \geq 1}$.

This is a decreasing sequence: $x^{n+1} = x x^n < x^n$.

Since $x > 0 \Rightarrow x^n > 0 \forall n \geq 1$.

Then $\lim_{n \rightarrow \infty} x^n = L$ (From convergence of monotone bounded sequences).

$$\begin{aligned} \text{Further } \lim_{n \rightarrow \infty} x^n &= \lim_{n \rightarrow \infty} x^{n+1} \quad \left[\overset{①}{(x, x^2, \dots)}, \overset{②}{(x^2, x^3, \dots)} \right] \\ &= \lim_{n \rightarrow \infty} x \cdot x^n \\ L &= xL \Rightarrow L(x-1) = 0 \\ \text{But } x \neq 1 &\Rightarrow L = 0. \end{aligned}$$

Thus for $0 < x < 1$, $\lim_{n \rightarrow \infty} x^n = 0$.

Story: We are talking about limits of sequences or convergence.

That is a sequence could be either convergent or divergent.

Is there something more to divergent sequences?

e.g. $a_n = (-1)^n (1 + 10^{-n}) \quad n \geq 1$

Claim: This sequence is not convergent.

Proof: Note that every convergent sequence is Cauchy.

Contrapositive: If a sequence is not Cauchy, it is not convergent.

$$\begin{aligned} \text{Then } |a_n - a_{n+1}| &= |(-1)^n [1 + 10^{-n} + 1 + 10^{-n-1}]| \\ &= |2 + 2 \times 10^{-n}| \\ &> 2 \quad (\text{For every } n). \end{aligned}$$

Choose $\epsilon = 1$, then $\forall n \geq 1 \quad \exists j = n, k = n+1$ s.t.

$$|a_j - a_k| > 2 > \epsilon$$

$\Rightarrow (a_n)_{n \geq 1}$ is not Cauchy and it is not convergent.

In particular $+1$ or -1 cannot be limits of $(a_n)_{n \geq 1}$.

$-1.1, 1.01, -1.001, 1.0001, \dots$ (This sequence doesn't seem to settle at one limit. However, it wants to settle at both ± 1 .)

Can we associate some meaning to ± 1 ?

Definition (Limit points):

Let $(a_n)_{n \geq 1}$ be a sequence of real numbers. Let $x \in \mathbb{R}$. Then

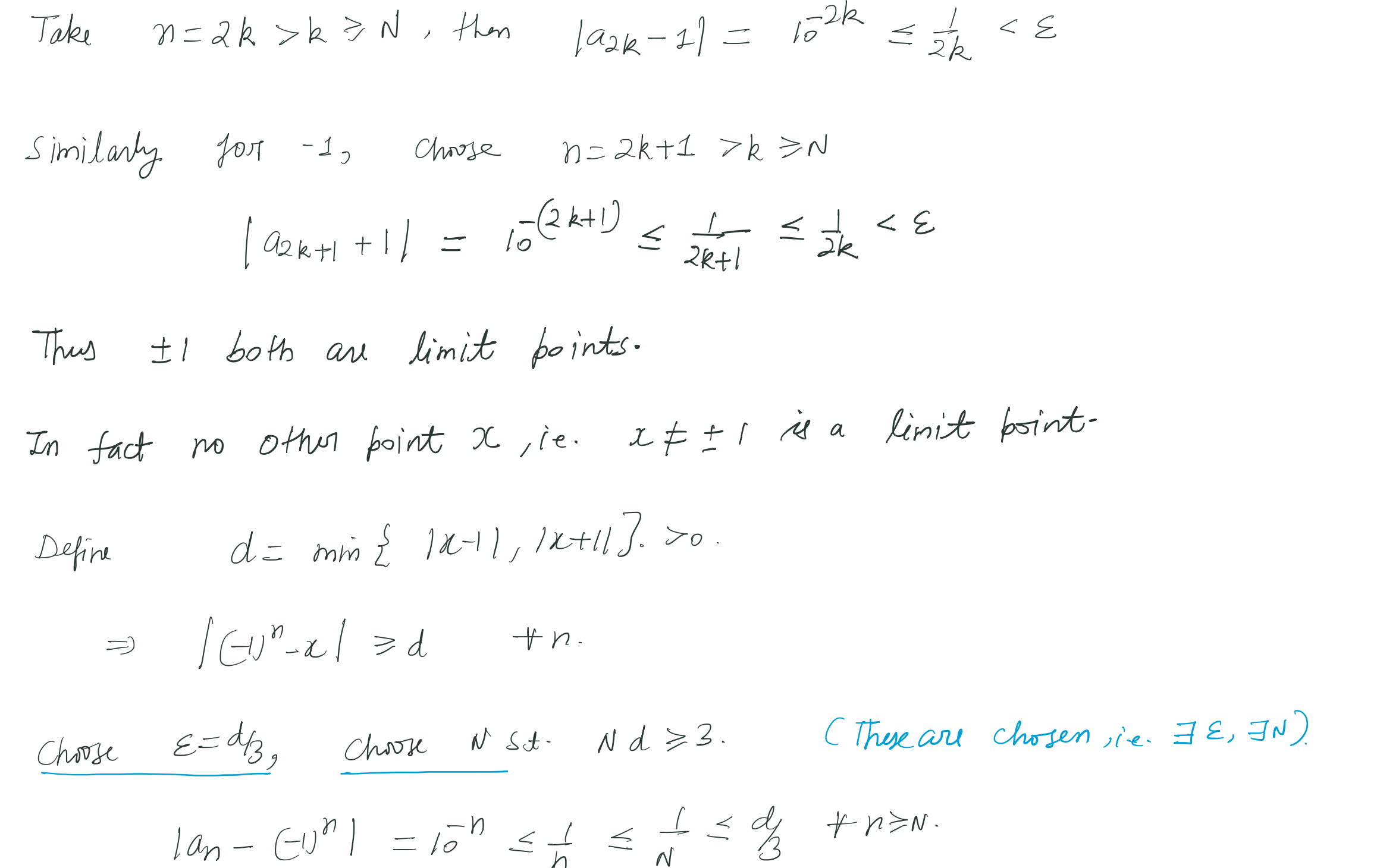
x is called a limit point of $(a_n)_{n \geq 1}$ iff

$$\forall \epsilon > 0 \quad \forall N \geq 1 \quad \exists n \geq N \quad \text{s.t.} \quad |a_n - x| \leq \epsilon.$$

Compare to convergence:

$$\forall \epsilon > 0 \quad \exists N \geq 1 \quad \forall n \geq N, \text{ we have } |a_n - x| \leq \epsilon.$$

Fix $\epsilon > 0$.



Ex: $a_n = (-1)^n [1 + 10^{-n}] \quad n \geq 1$. ± 1 both are limit points of $(a_n)_{n \geq 1}$.

Fix $\epsilon > 0$. Fix $N \geq 1$ as an arbitrary integer. Our goal is to

$$\text{find } n \geq N \quad \text{s.t.} \quad |a_n - 1| \leq \epsilon.$$

First note that $10^n \geq N \quad \forall n \geq 1$ (Induction).

Given N and ϵ , choose integer $k \geq N$ and $2 \leq k > 1$

$$\text{Take } n = 2k > k \geq N, \text{ then } |a_{2k} - 1| = 10^{-2k} \leq \frac{1}{2^k} < \epsilon$$

Similarly for -1 , choose $n = 2k+1 > k \geq N$

$$|a_{2k+1} + 1| = 10^{-(2k+1)} \leq \frac{1}{2^{k+1}} \leq \frac{1}{2^k} < \epsilon$$

Thus ± 1 both are limit points.

In fact no other point x , i.e. $x \neq \pm 1$ is a limit point.

$$\text{Define } d = \min \{ |x-1|, |x+1| \} > 0.$$

$$\Rightarrow |(-1)^n - x| \geq d \quad \forall n.$$

Choose $\epsilon = d/3$, choose N s.t. $Nd \geq 3$. (These are chosen, i.e. $\exists \epsilon, \exists N$)

$$|a_n - (-1)^n| = 10^{-n} \leq \frac{1}{n} \leq \frac{1}{N} \leq \frac{d}{3} \quad \forall n \geq N.$$

$$\text{Further } d \leq |(-1)^n - x| \leq |(-1)^n - a_n| + |a_n - x|$$

$$\leq d/3 + |a_n - x|$$

$$\Rightarrow |a_n - x| \geq 2d/3 > d/3 = \epsilon.$$

For this $\epsilon = d/3$, and this N , $\forall n \geq N \quad |a_n - x| > \epsilon$.

$$\Rightarrow x \text{ is not a limit point. } [\forall \epsilon \neq N \quad \exists n \geq N \quad |a_n - x| \leq \epsilon] \quad x \text{ being a limit point}$$

Claim (Limits are limit points): Let $(a_n)_{n \geq 1}$ be a sequence converging

to $L \in \mathbb{R}$. Then L is the only limit point of $(a_n)_{n \geq 1}$.

Proof: $\forall \epsilon > 0 \quad \exists N_0 \geq 1 \quad \text{s.t.} \quad \forall n \geq N_0 \quad |a_n - L| \leq \epsilon$.

Fix an $\epsilon > 0$. Let $\tilde{N} \geq 1$ be arbitrary.

$$\text{Define } N_1 = \max \{ N_0, \tilde{N} \}$$

$$\Rightarrow N_1 \geq \tilde{N}, \text{ and } N_1 \geq N_0 \Rightarrow |a_{N_1} - L| \leq \epsilon.$$

Thus for each $\tilde{N}$, there is N_1 s.t. $|a_{N_1} - L| \leq \epsilon$.

$$\Rightarrow L \text{ is a limit point.}$$

Suppose L' is also a limit point with $L' \neq L$.

$$\text{Let } \epsilon_0 = \frac{|L - L'|}{3}$$

$$\text{Given } \exists N_0 \geq 1 \quad \text{s.t.} \quad \forall n \geq N_0 \quad |a_n - L| \leq \epsilon_0. \quad (\text{Convergence})$$

$$\text{Further, } \exists N_1 \geq N_0 \quad \text{s.t.} \quad |a_n - L'| \leq \epsilon_0 \quad (\text{Limit point})$$

$$\Rightarrow |L - L'| \leq |L - a_n| + |a_n - L'|$$

$$\leq \frac{2|L - L'|}{3} \quad (\text{Contradiction})$$

Thus $L = L'$.

Limit superior and limit inferior:

Suppose $(a_n)_{n \geq 1}$ is sequence of real numbers. We define

two sequences $(a_n^+)_{n \geq 1}$ and $(a_n^-)_{n \geq 1}$ as follows.

$$\begin{aligned} a_n^+ &:= \sup_{n \geq n} a_n & a_n^- &:= \inf_{n \geq n} a_n \\ a_1^+ &= \sup \{ a_1, a_2, a_3, \dots \} & a_1^- &= \inf \{ a_1, a_2, a_3, \dots \} \\ a_2^+ &= \sup \{ a_2, a_3, \dots \} & a_2^- &= \inf \{ a_2, a_3, \dots \} \\ a_3^+ &= \sup \{ a_3, a_4, \dots \} & a_3^- &= \inf \{ a_3, a_4, \dots \} \end{aligned}$$

$$\text{Then } \limsup_{n \rightarrow \infty} a_n = \inf_{n \geq 1} a_n^+$$

$$\liminf_{n \rightarrow \infty} a_n = \sup_{n \geq 1} a_n^-$$

Ex: $a_n = (-1)^n [1 + 10^{-n}]$, what is $\limsup$ and $\liminf$ of $(a_n)_{n \geq 1}$.

Solution: Note that $a_{2k} = [1 + 10^{-2k}]$

$$a_{2k+2} = [1 + \frac{10^{-2k}}{100}] \leq a_{2k}$$

① So all even terms are positive and are decreasing.

$$a_{2k-1} = [1 + 10^{-2k} \times 10]^{x-1}$$

$$a_{2k+1} = [1 + 10^{-2k} \times \frac{1}{10}]^{x-1}$$

② so all odd terms are negative and are increasing.

Fix some $N \geq 1$. Then

$$a_N^+ = \sup \{ a_n : n \geq N \}$$

$$\text{If } N = 2k, \text{ then } a_N^+ \leq [1 + 10^{-2k}]$$

if we take any smaller number then that cannot be an upper bound as $[1 + 10^{-2k}]$ belongs to set.

$$N = 2k-1 \quad \text{then } a_N^+ \leq [1 + 10^{-2k}]$$

Same logic.

$$\text{Thus } N = 2k-1 \text{ or } 2k \quad a_N^+ = 1 + 10^{-2k}.$$

$$\Rightarrow a_1^+ = 1 + 10^{-2}$$

$$a_2^+ = 1 + 10^{-2}$$

$$a_3^+ = 1 + 10^{-4}$$

$$a_4^+ = 1 + 10^{-4}$$

$$\vdots$$

$$\inf_{N \geq 1} a_N^+ = \inf \{ 1 + 10^{-2k} : k \geq 1 \}$$

Since $1 + 10^{-2k}$ is decreasing and greater than 1

$$\inf_{N \geq 1} a_N^+ = \lim_{k \rightarrow \infty} [1 + 10^{-2k}] = 1$$

$$\Rightarrow \boxed{\limsup_{n \rightarrow \infty} a_n = 1.}$$

$$\text{Similarly, } N = 2k \text{ or } 2k-1 \quad a_N^- = -[1 + 10^{-(2k-1)}]$$

$$\sup_{N \geq 1} a_N^- = \sup \{ -(1 + 10^{-(2k-1)}) : k \geq 1 \}$$

This is increasing and upper bounded by $-1 \Rightarrow$

$$\lim_{k \rightarrow \infty} -[1 + 10^{-(2k-1)}] = \sup_{N \geq 1} a_N^-$$

$$\text{or } \sup_{N \geq 1} a_N^- = -1$$

$$\text{or } \boxed{\liminf_{n \rightarrow \infty} a_n = -1}$$